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SANDIP SINGH

Full Stack Engineer | Web3 Full Stack

PROFESSIONAL SUMMARY

Full-Stack & Blockchain Engineer with 5+ years building production web platforms, backend services and Web3 systems. Owns products end-to-end with React, Next.js, TypeScript, Node.js and PostgreSQL, and builds high-performance services and smart-contract integrations in Rust and Solidity. Recent work includes a payment and escrow platform that systematised ~95% of routine disputes, and a production booking/CRM platform for a UK consultancy.

TECHNICAL SKILLS

- **Languages:** TypeScript, JavaScript, Rust, Solidity
- **Frontend:** React.js, Next.js, Vite, Tailwind CSS, Radix UI, React Router, Responsive Web Design, Component-Based Architecture
- **Backend:** Node.js, Express.js, REST APIs, Serverless APIs, Middleware Architecture, API Validation, Webhooks, Rate Limiting, Error Handling, Audit Logging
- **Databases:** PostgreSQL, Supabase, MongoDB, Mongoose, Data Modelling, Database Migrations
- **Authentication & Security:** JWT Authentication, Role-Based Access Control (RBAC), Supabase RLS, API Security, Rate Limiting, Helmet, Consent Capture, Spam Protection, Honeypot Validation
- **Testing & Quality:** Vitest, Playwright, Unit Testing, Integration Testing, Smoke Testing, ESLint, Type Checking, npm Audit
- **DevOps & Deployment:** Docker, GitHub Actions, CI/CD, Vercel, Serverless Deployment, Environment Configuration, Production Deployment, Performance Optimisation
- **Additional Engineering:** Rust, Axum, Actix-web, Tokio, Microservices, Blockchain Integrations, Ethers.js, Viem, Web3.js

EXPERIENCE

Helson Software Solution Pvt Ltd – Full-Stack & Web3 Developer

08/2024 – 08/2026

UK Property-Survey Platform: nationalbuildingconsultants.co.uk

- Owned end-to-end delivery of a production platform (React, TypeScript, Supabase/PostgreSQL, Tailwind, Vercel), from architecture through deployment.
- Built customer workflows for enquiries, quotes, bookings and survey tracking, plus a CRM-style admin panel for customers, pricing, availability and report progress.
- Secured customer data with Row-Level Security, RBAC, server-side validation, rate limiting and audit logging; integrated Resend email and webhooks for booking lifecycle notifications.
- Set up CI/CD with GitHub Actions, Vitest, Playwright and ESLint for production-quality releases.

Blockchain Payment & Escrow Platform

- Built backend services, APIs and smart-contract integrations for a production payment and escrow platform processing 1,000+ blockchain transactions, covering settlement, pricing, disputes and monitoring.
- Automated ~95% of routine client-freelancer payment disputes through an AI-assisted workflow combining transaction history, intent analysis and business rules, with human review for edge cases.
- Cut pre-transaction latency by ~35% with a risk-analysis service in Rust (Actix-web), and built a low-latency pricing/oracle aggregation microservice using Axum and Tokio.
- Reached 90%+ test coverage across critical payment and settlement paths.

Gather Software Pvt Ltd –Blockchain Developer

01/2022 – 06/2024

- Architected and implemented distributed backend systems and blockchain infrastructure using Rust, JavaScript, Solidity, and Web3 technologies.
- Developed Rust-based protocol and application components for a custom Solana-based blockchain network running multiple validators.
- Built and maintained a dedicated Devnet environment for protocol, integration, and regression testing, shortening release cycles by 30%.
- Improved workflows for decentralised computing, task execution, pricing, transaction settlement, and platform monitoring, handling 10,000+ tasks/transactions.
- Built frontend integrations with Wagmi, Viem, and Web3.js connecting user-facing apps to backend and on-chain services.
- Developed indexing and analytics pipelines processing 100K+ on-chain transactions for reporting and monitoring.
- Blockchain & Web3: Solana, Smart Contracts, ERC-4337 (Account Abstraction), Ethers.js, Viem, Wagmi, Web3.js
- Systems & Performance: Rust, Axum, Actix-web, Tokio, Microservices, Distributed Systems

Sodio Technology Pvt Ltd –Jr. Blockchain Developer

04/2020 – 10/2021

- Built monitoring dashboards, backend services, and automation tools using JavaScript, Node.js, REST APIs, and Puppeteer.
- Developed automation tooling that reduced manual debugging effort by ~40% across a four-person engineering team.
- Supported smart-contract testing and blockchain data integrations for internal DApp projects.
- Collaborated across frontend, backend, and blockchain teams on requirement analysis, improving system reliability and usability.

EDUCATION

Master of Technology, NERIST, ARUNACHAL PRADESH.

07/2016 – 06/2018

7.74/10 GPA in Electronics and Communication.

Bachelor of Technology, SHIATS, PRAYAGRAJ, UTTAR PRADESH.

07/2012 – 06/2016

7.3/10 GPA in Electronics and Communication.

PUBLICATION

Sandip Singh, M. Edison, “A compact 3rd Order SIW Filter with Improved Upper Band Performance for Ku-Band Application”, International Conference on Innovation in Electronics, Signal Processing and Communication (IESC), IEEE 2019.